

EXECUTIVE SESSION
OCP GLOBAL SUMMIT 2026

OCT. 12, 2026

CHARACTERIZATION OF HOSE MATERIAL EFFECTS ON COOLANT CHEMISTRY DURING ACCELERATED AGING

AGENDA

01 Hose development for data center industry

02 Discuss need for advanced testing

03 The findings — preliminary, and what they suggest

04 Discuss changes and future testing

05 What it unlocks for the industry

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WHERE WE ARE AND WHERE WE ARE GOING

THE MATERIAL

- Peroxide-cured EPDM (pEPDM) is the preferred rubber for data center cooling hose
- Stable across a wide temperature range, durable under long heat exposure, and cured without metal activators or sulfur

WHAT WE SAW

- During permeation testing of various hose materials from different suppliers, we saw unusual fluid fouling
- There were even differences among pEPDM formulations

THE CONCERN

- Customer experience raised concern that long-term fluid–pEPDM interaction is under-studied, especially across the varied pEPDM formulations on the market
- The chemical interaction of coolant and the rubber tube material is important to understanding coolant degradation over time

OUR TEST

We used ASTM D471 accelerated aging to test different hose tube materials' long-term effects on common coolants (DI-water based fluids)

8 weeks at 80°C
Artificial aging of pEPDM hoses

1m hose samples (-8 size)
Real-world conditions minus flow

Test controls to prevent fluid leaks
Maximize surface area for molecule exchanges

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WHY PEROXIDE-CURED EPDM FOR DATA CENTER COOLING HOSES?

THE INCUMBENT MATERIAL

Peroxide-cured EPDM meets industry requirements. That is why it became the default.

PEROXIDE-CURED EPDM

- ✓ **Metal-free cure system**
- ✓ **Peroxide cure package does not add metal ions**
- ✓ **Non-conductive, ozone-resistant**
- ✓ **Handles processor temperature range**
- ✓ **High flexibility (routes through tight rack geometry) with exemplary durability**

SULFUR-CURED EPDM

Sulfur-cured EPDM is the standard in most industries but is not suitable for data centers.

- ✗ **Zinc required for activation in cure package**
- ✗ **Produces metal ions**

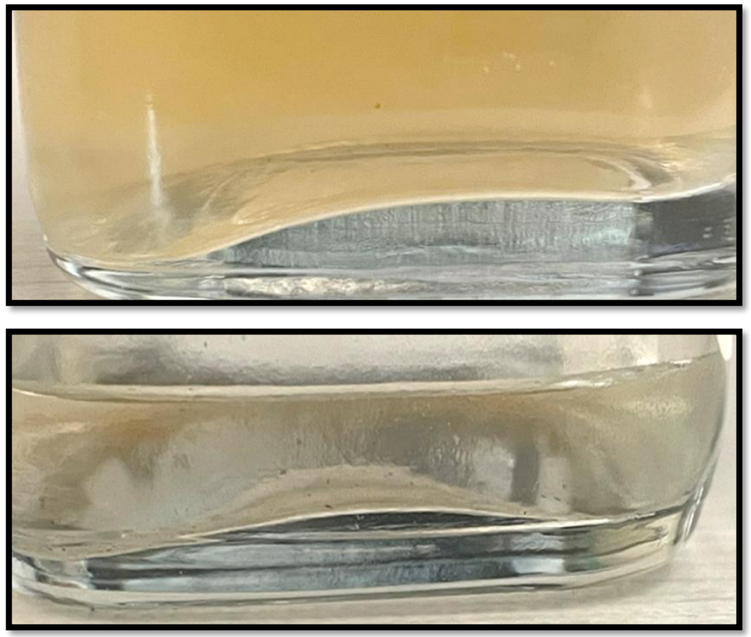
THE DEFAULT TUBE MATERIAL, NOW UNDER THE MICROSCOPE

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THEN THE COOLANT STARTED CHANGING COLOR

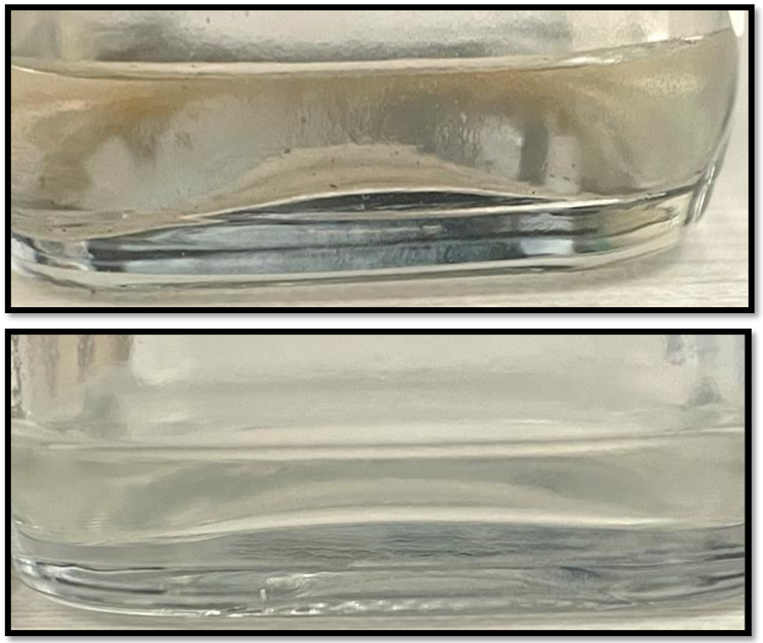
WHAT WE SAW

- Some coolant fouled during permeation testing
- Brownish-orange tinge, suspended particles



WHAT WAS IN THE TEST

- Multiple manufacturers
- Several tube materials, including peroxide-cured EPDM



WHAT WE DID NOT KNOW

- Source of the fluid fouling
- Why did fluid soaked in different pEPDM hoses have vastly different colors?
- Are rubber formulations really so different from each other in this industry?
- Hypothesis
  - Possible rubber dust contamination and leaching from the rubber tube materials

AN UNPLANNED OBSERVATION BECAME THE REASON FOR THIS STUDY

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UNDERSTANDING FLUID DEGRADATION IS IMPORTANT

- **Cooling fluids are an engineered combination of chemicals**
  - De-ionized water for optimum heat transfer
  - Propylene Glycol to expand the operating temperature range
  - Corrosion inhibitors
  - Biocides to prevent microbial growth
- **Reduced heat-transfer performance due to restricted flow at the cold plate or within the system pumps**
  - Customers have reported polymerization of their cooling fluids during testing (slimy strands)
- **Corrosion and erosion from materials introduced by the hose tube**
- **Any of these issues reduce performance of the data center and every issue scales with the number of components within a cooling loop — cooling system failure is possible if the issue is not addressed**

CLEAN FLUID PATHS PROTECT THERMAL PERFORMANCE, FLOW EFFICIENCY AND SYSTEM RELIABILITY

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WHAT WE ARE SEEING SO FAR

RESULTS AT A GLANCE: WHAT THE FLUID TOLD US

Phase 1 accelerated aging with DI water (8 weeks at 80° C). Each result is read against a control from the same coolant batch.

INVOLATILE RESIDUE IN FLUID (PPM)

Control fluid

- Na, Si, K dominant
- Al, P, S, Cl, Ca, Cu present in smaller quantities

Sulfur-cured EPDM

- Fatty stearates, accelerator fragments, zinc dibutyl di thiophosphate, and butanol from tube
- Na, P, S, Zn and other chemicals from contaminants

Peroxide-cured EPDM #1

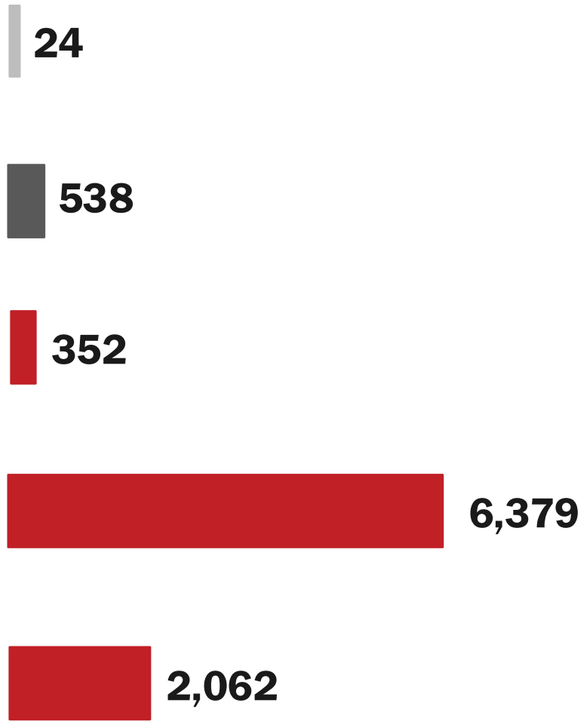
- Dioctyl terephthalate and ethyl hexanol from tube
- Na, Si, S, Cl, K, Ca, Zn from contaminants and likely also extracted from tube

Peroxide-cured EPDM #2

- Dioctyl terephthalate and ethyl hexanol from tube
- Fe, Zn oxides from contamination

Peroxide-cured EPDM #3

- Tributyl phosphate from bulk tube
- Na, Mg, Si, P, S, Cl, K, Ca contaminants and likely also extracted from tube



KEY TAKEAWAYS

1 Sulfur-cured EPDM performed as expected

- It leached zinc, sulfur and phosphorus, which confirms why it is not suitable for data centers

2 Not all pEPDM is the same

- Residue across the three pEPDM samples ranged from 352 to 6,379 ppm
- Formulation and cleanliness both matter
- In one hose sample, it is likely the plasticizer is coming out of the rubber and breaking down in the coolant

3 Contamination was a major factor

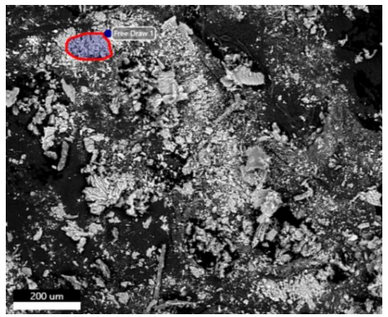
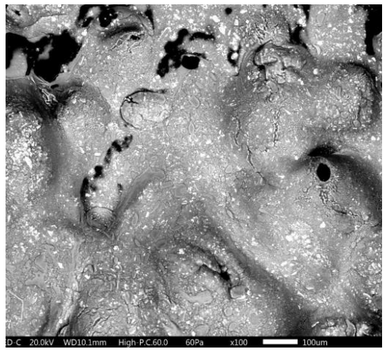
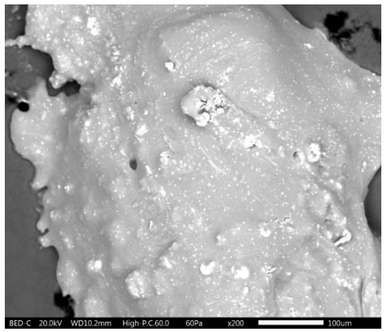
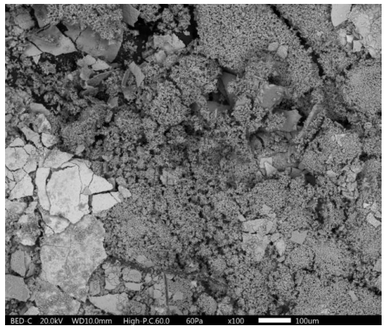
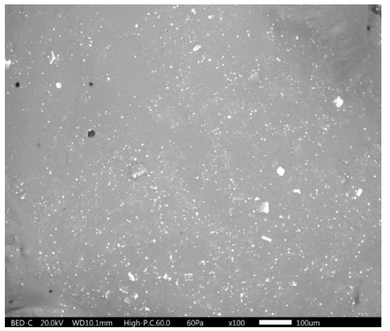
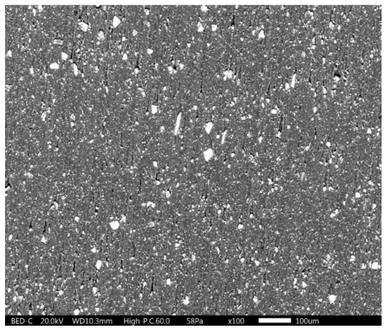
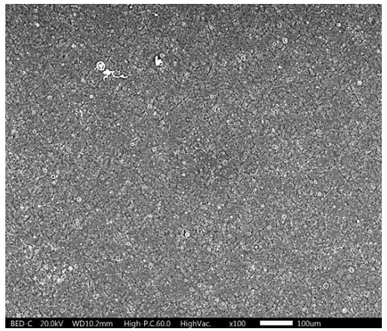
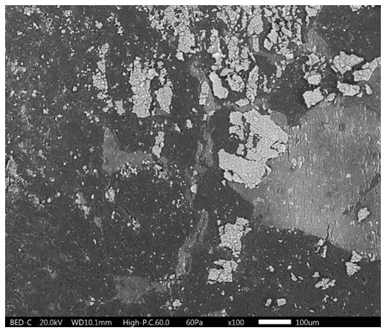
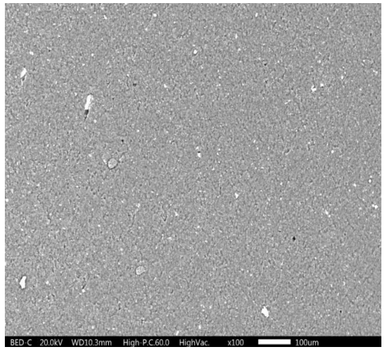
- Metals in the pEPDM fluids most likely came from manufacturing contaminants and lab DI water
- Phase 2 adds hose rinsing, DI water from a different source, and tighter controls on leak detection

PEROXIDE-CURED EPDM IS STILL THE RIGHT CHOICE, BUT FORMULATION AND CLEANLINESS DRIVE RESULTS

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RESULTS AT A GLANCE: TEST IMAGES

Phase 1 accelerated aging (8 weeks at 80úC). Each result is read against a control from the same coolant batch.

	Control fluid 24 ppm	Sulfur-cured EPDM 538 ppm	Peroxide-cured EPDM #1 352 ppm	Peroxide-cured EPDM #2 6,379 ppm	Peroxide-cured EPDM #3 2,062 ppm
SEM: DRIED PARTICULATES					
SEM: HOSE TUBE SURFACE	No hose (fluid only)				

MULTI-METHOD ANALYSIS PINPOINTS METAL OXIDE CONTAMINATION DOWN TO THE TUBE SURFACE

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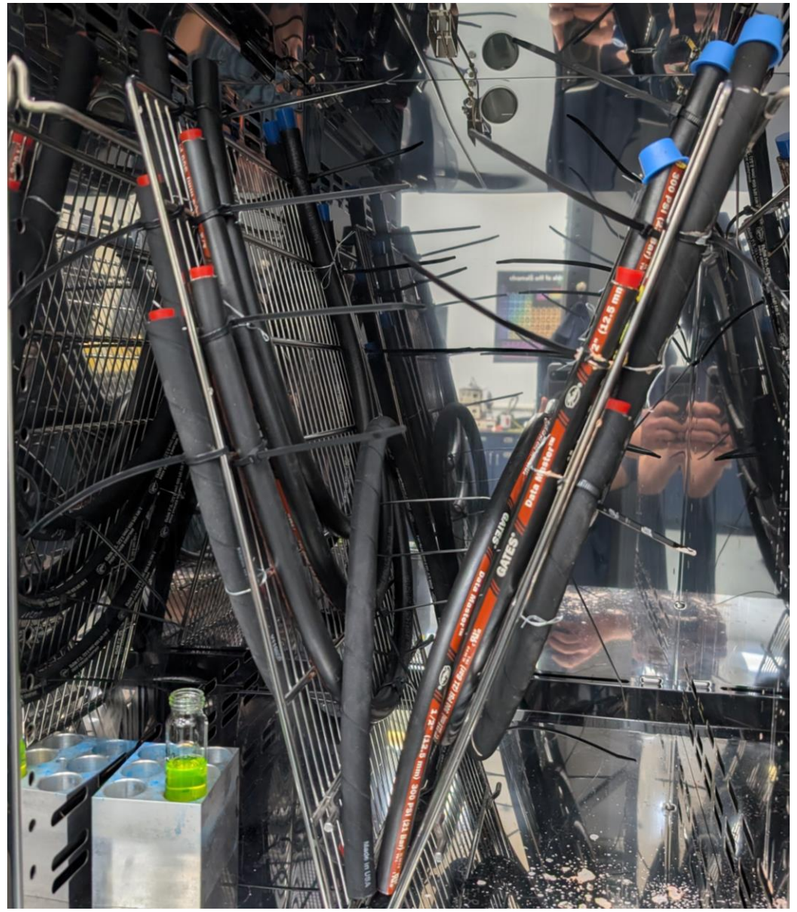
ADDITIONAL TESTING

MORE TESTING TO REACH CONCLUSIONS

METHOD AND CONDITIONS

Gates method, based on ASTM D471

1. Rinse hoses; air dry in a controlled environment
2. Add fluid to hoses
3. Weigh and cap filled hoses
4. Age at 80 °C for 1,344 hours (8 weeks)
5. Check periodically for leaks (visual, weight)
6. Uncap and weigh hoses
7. Collect fluids in sterile containers
8. Analyze



WHAT GETS MEASURED

- Mass loss: hose and fluid, before/after
- Fluid chemistry: GC-MS on residue
- Fluid involatile residue: SEM microscopy
- Hose tube surface microscopy
- Hose tube cross-section microscopy

CONTROLS

- Control fluid: same source as samples
- Control hoses: aged, unsoaked

EVERY RESULT IS READ AGAINST A CONTROL FROM THE SAME COOLANT OR HOSE BATCH

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RESULTS IN PROGRESS, SHOWN AS DIRECTIONAL ONLY

- **Initial test results show an issue with manufacturing contamination and a need for better test controls**
- **Phase 1 of testing was concluded as contamination became a concern**
  - The results have already guided us on additional concerns and how to investigate them
- **Phase 2 testing will improve upon the first phase test design by reducing contamination, controlling fluid leakage, and providing a better baseline of material exchange**
  - Phase 2 testing began in September
  - Five different pEPDM samples have already been prepared
- **Phase 3 testing will use material coupons in fluid near boiling temperatures instead of soaking fluid in tubes**
  - This test will introduce control methods to prevent evaporation of fluid
- **Development of chemical baselines and knowing how the fluid chemistry changes when in contact with hose tube materials will help develop real-time fluid monitoring systems to prevent costly maintenance during data center operation**

DIRECTIONAL, NOT FINAL. NUMBERS WILL MOVE AS THE REMAINING SAMPLES COMPLETE

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SUMMARY AND CALL TO ACTION

SUMMARY

- **Peroxide-Cured EPDM is still the right material for data center cooling hoses, but material formulations and cleanliness vary widely, leading to varied results**
- **Extractables in the fluid can be impactful on the overall liquid cooling system**
  - Reduced heat-transfer performance
  - Restricted flow
  - Pump degradation
  - Plogging issues
  - Corrosion
- **Clean fluid paths protect thermal performance, flow efficiency and system reliability**

CALL TO ACTION

- **Collaborate with us on this topic**
  - Suggestions for modifications or results you'd like to see
  - Share similar results for comparative analysis
  - Seeking collaborators for a cleanliness spec or sub-project
- **Where to find additional information**
  - Data Center Cooling Solutions
  - Visit the Gates Corporation in Hall 3, Booth #D32
  - Scan the QR code on the next slide to connect with us for questions, comments, or collaboration interest

SHARING OUR RESULTS TO HELP DEVELOP BETTER ADVANCED WARNING IN OUR FLUID MONITORING SYSTEMS

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